Actionability Energy: A Tangent-Space Test for Controllable Visual Futures

Anonymous submission-ready draft

Abstract

Generated futures are often evaluated by visual plausibility, passive likelihood, or goal reward. For embodied control, these criteria are incomplete: a future can look plausible while requiring instantaneous state changes outside the local action-reachable set of the robot. We formalize this failure mode with an *actionability residual*, the distance between an imagined representation transition and the tangent space induced by an embodiment action map. The residual is used both as a diagnostic score and as a composable factor in trajectory inference. We introduce a low-compute benchmark with a holonomic point mass, nonholonomic car, two-link arm, and contact-switching planar pusher. Across these settings, the residual predicts execution failure better than smoothness, passive-prior, path-length, and action-norm baselines. Adding the residual to a repair objective converts non-executable generated futures into trajectories that are locally executable under the corresponding embodiment. The study is deliberately controlled: it does not claim real-robot validation or large video-model finetuning. Its contribution is to isolate a representation-level criterion for asking whether an imagined future is locally controllable by the body that must execute it.

1 Introduction

World models and video planners give embodied agents a tempting interface: imagine futures, score them against a goal, and execute the best one. This interface is insufficient when the future is evaluated only as a passive visual sequence. A generated transition may be smooth, likely under video data, and goal-satisfying, while still being impossible for the current embodiment to realize in one local step. A nonholonomic car cannot move sideways without changing heading. A pusher cannot move an object before contact. An arm near a singular configuration cannot realize every Cartesian displacement within its action limits.

This paper studies the local test that is missing from such futures. Let z_t be a representation state and let $J_\phi(z_t)$ map small controls into representation-space changes. The local action-reachable tangent set is

$$\mathcal{A}(z_t) = \{J_\phi(z_t)u : u \in \mathcal{U}\}.$$

For an imagined transition $\Delta z_t = z_{t+1} - z_t$, we define the actionability residual

$$r_t = \min_{u_t \in \mathcal{U}} \|\Delta z_t - J_\phi(z_t)u_t\|_W^2 + \lambda \|u_t\|^2.$$

The score is small when the transition can be explained by a bounded local action and large when the generated future leaves the embodiment's reachable tangent set.

The idea is complementary to recent Jacobian-based robot policies and executable-video systems. Neural Jacobian Fields and VERA show that local visual Jacobians can translate desired visual change into action [7, 8]. EVA aligns video world models with inverse-dynamics rewards [12]. World Action Verifier asks whether world-model futures are executable through forward-inverse asymmetry [10]. Here we isolate a simpler representation-level question: before or during trajectory inference, can we certify that the local transition lies near the action tangent set of the current embodiment?

Contributions. First, we formalize an actionability residual for generated representation trajectories and connect it to local execution error. Second, we build a CPU-first benchmark with four toy embodiments that expose different non-executability modes. Third, we show that the residual predicts execution failure and supports repair when composed with world, goal, smoothness, and constraint energies. Fourth, we demonstrate embodiment swapping: the same imagined displacement can be feasible for one body and infeasible for another by changing only J_ϕ.

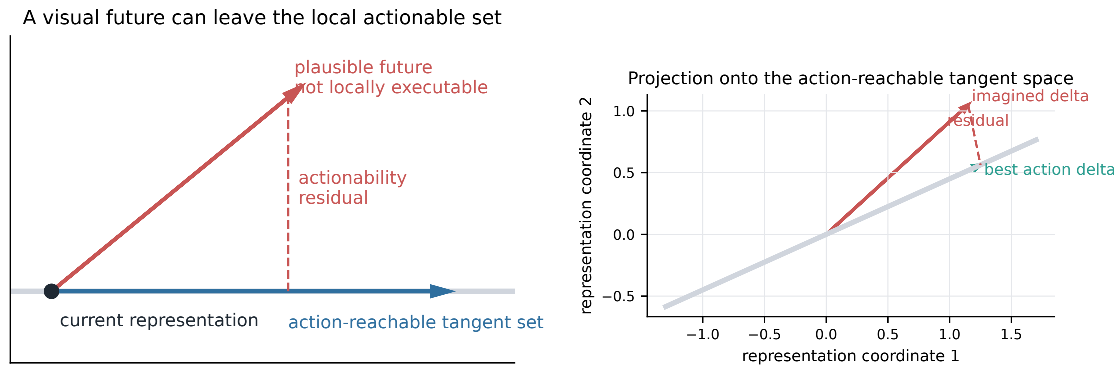

Figure 1: A generated future can be visually plausible while leaving the local action-reachable tangent set. The actionability residual projects an imagined representation change onto $\mathrm{Im}(J_\phi)$ under action limits and penalizes the residual.

2 Related Work

Jacobian representations for robot action. Neural Jacobian Fields learn visual Jacobian fields that map desired visual changes to robot actions across diverse robots [7]. VERA uses a video planner with a Jacobian-based inverse dynamics module to produce generalist robot policies [8]. Our work does not improve NJF or reproduce VERA-scale robot experiments. It uses the same kind of local action-sensitivity object for a different purpose: diagnosing whether a generated transition is locally executable and repairing trajectories at inference time.

Executable video and world-action verification. EVA aligns video world models with executable robot actions through inverse-dynamics rewards [12]. World Action Verifier studies self-improving world models through forward-inverse asymmetry [10]. These works are the closest novelty boundary. Our residual is not a learned verifier loop or a large-video-model finetuning objective. It is a low-compute tangent-space certificate that can be computed from analytic, finite-difference, or small learned action maps.

Energy-based planning and compositional inference. Energy-based planning frames control as inference over composed objectives [5]. LEAP performs iterative energy minimization with sequence models [2], and compositional generative modeling studies how separate energy factors can be combined at inference time [4]. We use actionability as one factor in such a composed trajectory energy, rather than as a decorative name.

Queryable scene and world representations. Neural scene representations for visuomotor control demonstrate that learned representations can support robot control queries rather than only rendering [9]. RoboDreamer and Large Video Planner push world-model imagination toward robot planning [1, 13]. Our claim is narrower: a representation for embodied planning should expose local actionability, not only visual prediction or geometric state.

Failure modes in passive learning. Visual foresight and model-predictive control showed that learned video prediction can be useful for manipulation [6]. Causal confusion and observational overfitting show that passive correlations can mislead policies or value estimates [3, 11]. Actionability is a related embodied failure mode: a passive future can be plausible but physically unreachable for the current body.

3 Actionability Energy

3.1 One-Step Residual

Given a representation transition $\Delta z_t = z_{t+1} - z_t$ and local action map $J_\phi(z_t) \in \mathbb{R}^{d_z \times d_u}$, the unconstrained residual has the closed form

$$u_t^\star = (J^\top W J + \lambda I)^{-1} J^\top W \Delta z_t,$$
$$r_t = \|\Delta z_t - J u_t^\star\|_W^2 + \lambda \|u_t^\star\|^2.$$

When actions are box-bounded, $u \in [u_{\min}, u_{\max}]$, the implementation solves the same quadratic program under bounds. Because the benchmark action spaces are one- and two-dimensional, we use an exact active-set enumeration for speed and keep a generic SciPy optimizer fallback for larger action spaces.

3.2 Trajectory Energy

For a candidate future $\tau = z_{0:T}$ and goal g, we compose

$$E(\tau) = E_{\text{world}}(\tau) + \alpha E_{\text{goal}}(\tau, g) + \beta \sum_{t=0}^{T-1} r_t + \gamma E_{\text{smooth}}(\tau) + \eta E_{\text{constraints}}(\tau).$$

E_{world} keeps the repaired trajectory near the proposed future, E_{goal} pulls the terminal state toward the task target, and $\sum_t r_t$ penalizes non-actionable local transitions. In this package, repair is implemented as projected trajectory inference: build a reachable rollout from local action projections or simple embodiment-aware controllers, then blend it with world and goal terms. This is intentionally CPU-first and should be read as a benchmark solver, not as a claim of optimal nonlinear trajectory optimization.

3.3 Embodiment Swapping

The same representation displacement can be paired with different J_ϕ maps. A sideways (x, y) displacement has zero residual for a holonomic point mass but high residual for a car whose heading points forward. A pusher-object displacement is actionable only under contact. This makes actionability a property of a representation plus an embodiment, not a property of the visual future alone.

4 Theory: Plausibility Is Not Executability

Proposition 1 (Passive plausibility does not imply actionability)**.** *There exist two trajectories τ_a, τ_b with equal smoothness and equal terminal goal reward under a passive trajectory score, such that τ_a is locally actionable and τ_b has strictly positive actionability residual.*

Proof. Consider a nonholonomic car at $z_0 = (0, 0, 0)$ with representation (x, y, θ) and local Jacobian

$$J(z_0) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

after absorbing the time step into units. Let τ_a use $\Delta z = (\epsilon, 0, 0)$ and τ_b use $\Delta z = (0, \epsilon, 0)$ for $\epsilon > 0$. Both one-step trajectories can have identical Euclidean length and identical passive smoothness, and a goal-only score can assign either transition the preferred terminal reward by placing the goal at the corresponding endpoint. For τ_a, $\Delta z \in \operatorname{Im}(J)$, so the residual can be zero up to the ridge term. For τ_b, the y component is orthogonal to $\operatorname{Im}(J)$, so the residual is at least ϵ^2 for $W = I$ and $\lambda = 0$. Thus passive plausibility and goal reward alone do not certify actionability. □

Proposition 2 (Residual lower-bounds local execution error)**.** *Assume local dynamics in representation space satisfy*

$$z_{t+1} = z_t + J(z_t) u_t + \rho(u_t), \quad \|\rho(u_t)\| \le c \|u_t\|^2$$

for bounded u_t. Let $r_t^0 = \min_{u \in \mathcal{U}} \|\Delta z_t - J(z_t) u\|_2^2$ be the unregularized residual. Then any one-step execution attempting to realize Δz_t has representation error at least $\sqrt{r_t^0} - c\|u_t\|^2$.

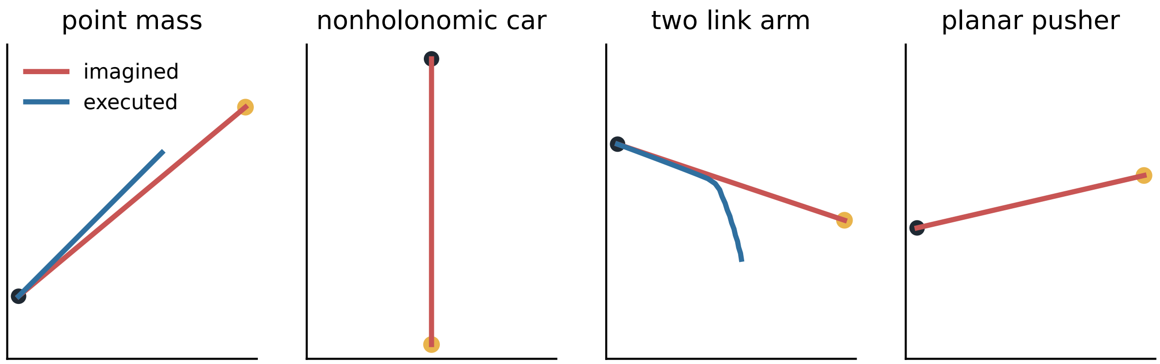

Figure 2: Representative imagined trajectories and rollouts under recovered controls. The environments isolate different reasons a plausible representation future can fail under the executing body.

Proof. For any feasible u_t,

$$\|\Delta z_t - (Ju_t + \rho(u_t))\|_2 \geq \|\Delta z_t - Ju_t\|_2 - \|\rho(u_t)\|_2.$$

By definition, $\|\Delta z_t - Ju_t\|_2 \geq \sqrt{r_t^0}$ for every feasible action. Applying the second-order bound on ρ gives the result. The bound is local: if actions are large or contact modes switch, the second-order term can dominate. $\square$

These propositions justify using the residual as a diagnostic rather than as a complete dynamics model. The residual detects local tangent-set mismatch; it does not prove long-horizon feasibility under arbitrary nonlinear dynamics.

5 Low-Compute Benchmark and Experiments

Environments. The benchmark uses four deterministic CPU environments. The point mass is a positive control with $J = I$. The nonholonomic car exposes sideways visual futures that are locally impossible. The two-link arm uses an analytic end-effector Jacobian and action limits. The planar pusher has a contact-dependent Jacobian, so object motion before contact is non-actionable.

Generated futures. We use controlled future generators rather than large video models: straight-line interpolation, noisy splines, passive smoothness priors, scripted rollouts, and intentionally non-actionable futures. These generators are proxies for generated visual futures and are not claimed to match real video-model distributions.

Experiment 1: constructive counterexample. The car example in Proposition 1 is implemented directly. A sideways future has the same endpoint smoothness as a forward future under passive scoring, but the residual rejects it because the displacement lies outside the car's local tangent set.

Experiment 2: failure prediction. For each candidate, we compute residuals, recover controls, roll out the true environment, and mark failure when the executed path diverges from the imagined path. Table 1 reports the actionability score. In the saved CSVs, it is compared to smoothness, path length, action norm, passive negative log-likelihood proxy, and random score.

Experiment 3: repair. We compare the original candidate, goal-only, world+goal, world+goal+smoothness, world+goal+actionability, and full energy variants. Adding actionability changes the trajectory rather than only rescoring it; it encourages futures whose local increments can be produced by bounded actions.

Table 1: Failure prediction with the actionability residual. Higher scores predict execution failure.

Environment	AUROC	AUPRC	Acc.
nonholonomic car	1.000	1.000	0.900
planar pusher	1.000	1.000	0.900
point mass	1.000	1.000	0.900
two link arm	0.990	0.966	0.811
all	0.989	0.989	0.917

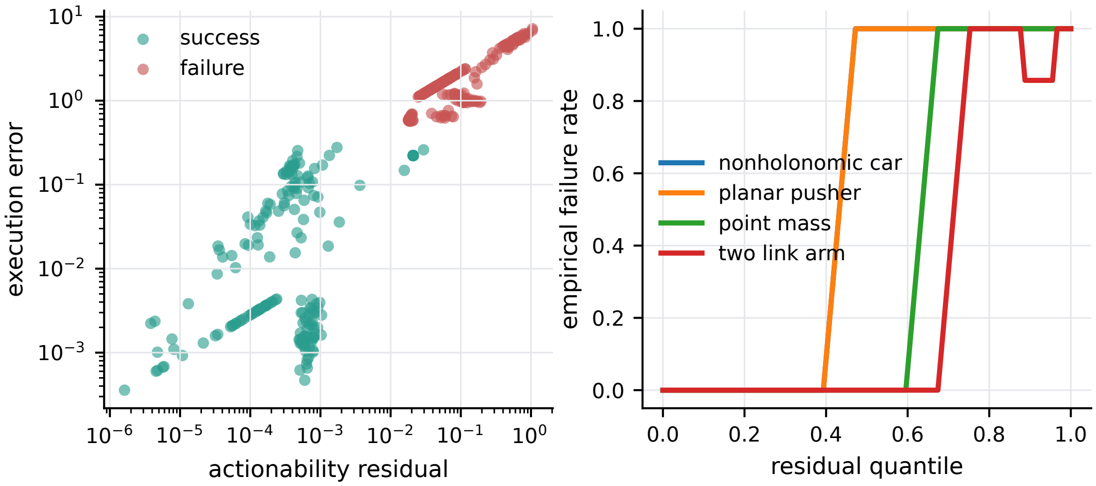

Figure 3: Actionability residuals correlate with rollout error and produce monotone failure-rate curves across embodiments.

Experiment 4: embodiment swap. We score the same local (dx, dy) displacement under different action maps. The residual changes sharply across embodiments, showing that executability is body-conditioned.

Experiment 5: learned J_ϕ. We fit small local linear action maps from random rollouts and compare them to analytic Jacobians. Error decreases with sample count, but the pusher remains harder because contact discontinuities violate a single smooth local model.

6 Limitations

The benchmark is toy-only and deterministic. It tests local representation-space feasibility, not real robot reliability. The repair solver is a CPU projected-inference routine, not a full nonlinear optimal-control solver. The residual is local and can be misleading when second-order effects, contact discontinuities, perception errors, or long-horizon compounding dominate. Learned Jacobians are evaluated with small local regressors rather than large neural fields. The pusher exposes mode switching, but it is not a physically faithful contact simulator. These limitations are intentional for a first low-compute diagnostic paper, but real robot validation would be required before making claims about deployment.

7 Conclusion

Actionability energy formalizes a simple requirement for embodied world models: imagined representation changes should lie near the local action-reachable tangent set of the body that must execute them. In controlled CPU benchmarks, this residual predicts execution failure, supports inference-time repair, and changes under embodiment swaps. The result is not a replacement for video planners, inverse dynamics, or

Table 2: Repair success on diagnostic non-executable futures. The full energy includes world, goal, smoothness, and actionability terms.

Environment	Candidate	Full energy
nonholonomic car	0	1
planar pusher	0	1
point mass	0	1
two link arm	0	1

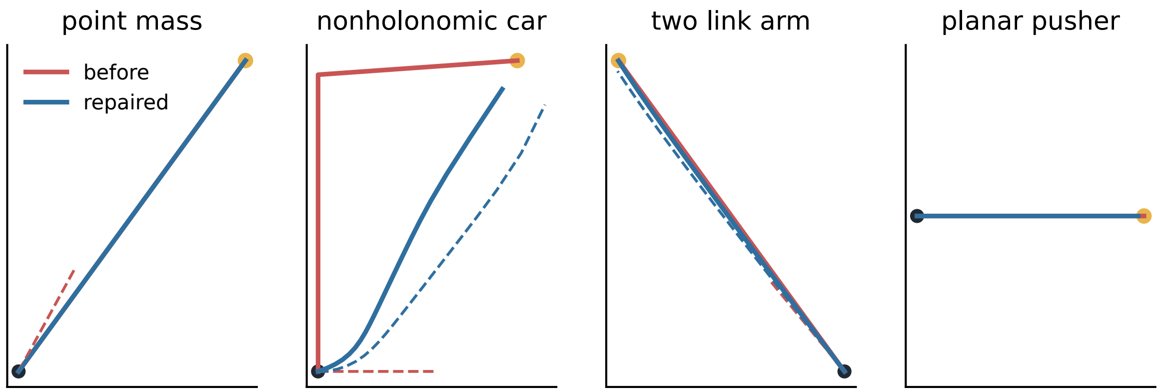

Figure 4: Before/after repair examples. Solid lines show imagined futures; dashed lines show rollouts from recovered controls.

real robot evaluation. It is a diagnostic and repair factor that can make generated futures more honest about what an embodiment can locally do.

8 Reproducibility Statement

All experiments are CPU-first Python scripts. The repository includes environment code, residual solvers, repair routines, tests, raw CSV results, plotting scripts, and LaTeX source. The intended verification commands are:

```
pytest -q
python experiments/run_all.py --quick
python experiments/run_all.py --full
bash scripts/make_plots.sh
bash scripts/compile_paper.sh
```

The generated reports record command status, dependency versions, hardware, unresolved issues, and readiness level.

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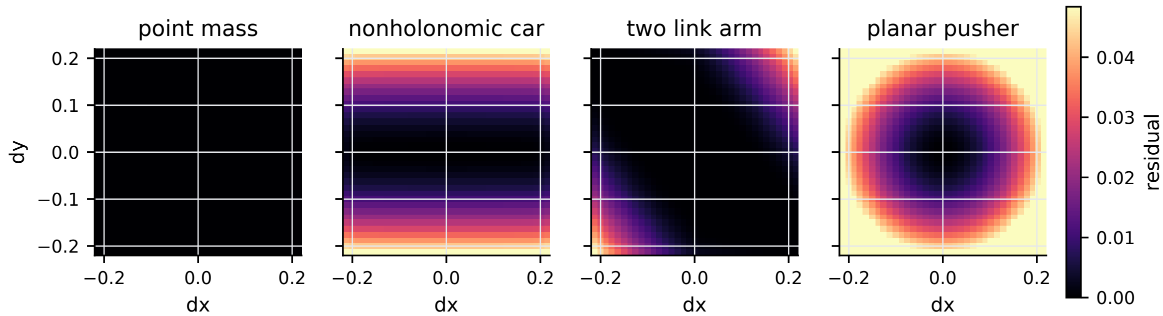

Figure 5: Embodiment swap heatmaps. The same representation displacement can be actionable for one body and non-actionable for another.

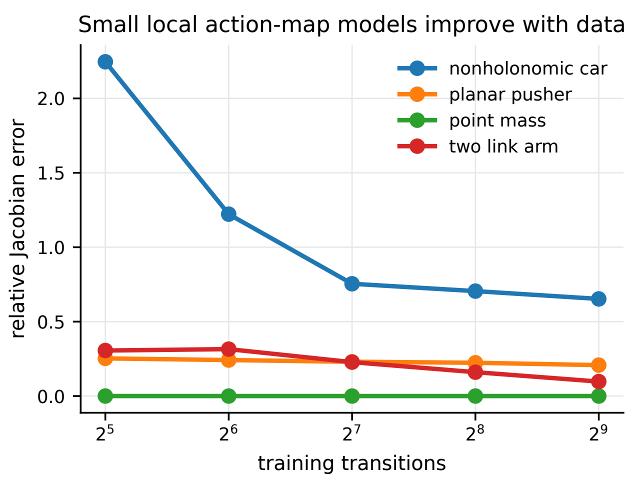

Figure 6: Sample efficiency of small local action-map models.

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A Additional Implementation Details

Residual solver. For unconstrained actions, the solver uses the ridge projection closed form. For bounded actions, the benchmark uses exact active-set enumeration over low, free, and high states for each action coordinate. This is exact for the convex quadratic objective in the low-dimensional action spaces used here.

Candidate generators. Candidates include true rollouts, straight-line paths, too-fast point-mass jumps, sideways car paths, Cartesian arm teleports, and object-motion-before-contact pusher paths. Labels are not assigned by construction; they are produced by recovering controls from the candidate and rolling out the environment.

Repair solver. The repair routine constructs a reachable trajectory from local action projections or embodiment-aware controllers and blends it with world, goal, and smoothness terms. This is a practical CPU solver for the benchmark. Future versions should compare against differentiable nonlinear trajectory optimization and learned video latent repair.

Novelty boundary. The closest overlap found is World Action Verifier [10]. The present work should avoid claiming to be the first executable-video verifier. Its narrower claim is a tangent-space actionability residual, benchmark, and repair energy for local embodiment-conditioned generated futures.